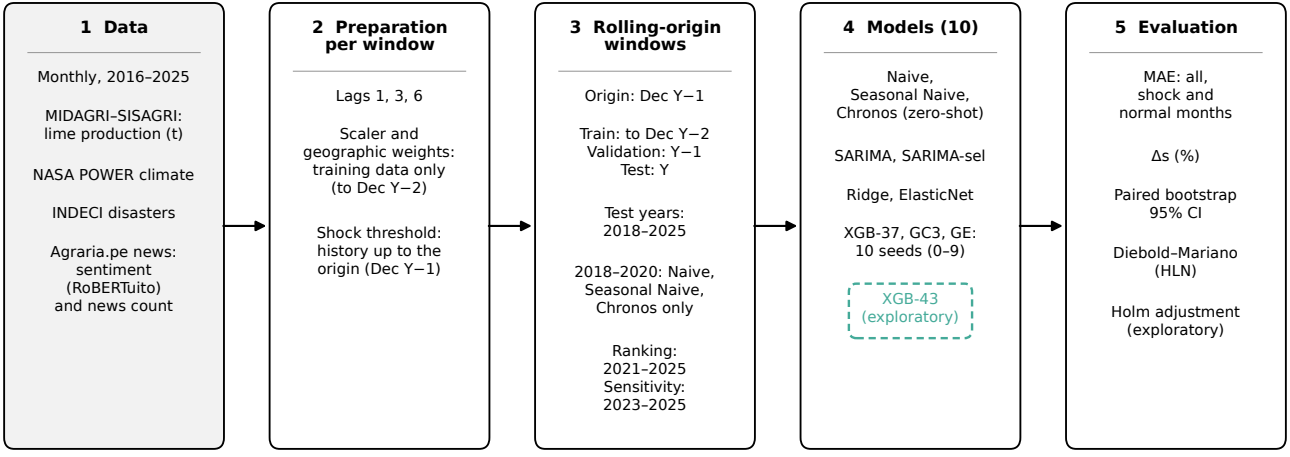
1 Data
Monthly, 2016–2025
MIDAGRI–SISAGRI: lime production (t)
NASA POWER climate
INDECI disasters
Agraria.pe news: sentiment (RoBERTuito) and news count
2 Preparation per window
Lags 1, 3, 6
Scaler and geographic weights: training data only (to Dec Y−2)
Shock threshold: history up to the origin (Dec Y−1)
3 Rolling-origin windows
Origin: Dec Y−1
Train: to Dec Y−2
Validation: Y−1
Test: Y
Test years: 2018–2025
2018–2020: Naive, Seasonal Naive, Chronos only
Ranking: 2021–2025
Sensitivity: 2023–2025
4 Models (10)
Naive, Seasonal Naive, Chronos (zero-shot)
SARIMA, SARIMA-sel
Ridge, ElasticNet
XGB-37, GC3, GE: 10 seeds (0–9)
XGB-43 (exploratory)
5 Evaluation
MAE: all, shock and normal months
Δs (%)
Paired bootstrap 95% CI
Diebold–Mariano (HLN)
Holm adjustment (exploratory)